

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 MATHEMATICS EXTENSION 1

PRELIMINARY ASSESSMENT TASK 2

JULY 2006

Time allowed: 70 minutes

Instructions:

- Show full working (this is important, especially in “show that” questions)
- Start each question on a new page
- Full marks may not be awarded for careless or badly arranged work
- Non-programmable calculators may be used
- This paper must be handed in on top of your answer sheets
- Answers must be written in blue or black pen
- Answers must be arranged in order and stapled securely. No claims for missing pages will be allowed.

Name: _____

Class: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Total
/11	/11	/11	/11	/11	/55

Question 1

- (a) Evaluate
- i) $\lim_{x \rightarrow \infty} \frac{x^2 + 3x + 1}{2x^2 - 4}$ 1
- ii) $\lim_{x \rightarrow 3} \frac{x^2 + 3x - 18}{x^2 - 5x + 6}$ 1
- (b) Consider the interval joining the points A(1, -4) and B(5, 12).
- i) Find the point which divides the interval AB externally in the ratio 3:1 2
- (ii) Find the ratio in which the x axis divides the interval AB. 1
- (c) Show on the Cartesian plane the union of the regions represented by the following inequalities: $y < x - 2$, $y \geq 0$ & $x \leq 0$. 3
- (d) The acute angle between the lines $y = 3x + 5$ and $y = mx + 4$ is 45° . 3
Find the possible values of m .

Question 2 (Start a new page)

- (a) Using the notation $\frac{d}{dx} f(x) = f'(x)$ and $\frac{d}{dx} g(x) = g'(x)$, write in simplest form: 4
- i) $\frac{d}{dx} [f(x) + g(x)]$
- ii) $\frac{d}{dx} [f(x) \cdot g(x)]$
- iii) $\frac{d}{dx} \frac{1}{f(x)}$
- iv) $\frac{d}{dx} f(g(x))$

Question 2 (cont.)

- (b) For the curve. $y = \frac{x+1}{x-1}$ 7
- i) The curve cuts the x axis at P. Find the x value at P.
 - ii) Find $\frac{dy}{dx}$.
 - iii) Find the gradient of the tangent to the curve at P.
 - iv) Using the point gradient form of a straight line, show how the equation of the normal at P, $y = 2x + 2$, may be derived.
 - v) State clearly the x value of the point where the normal again cuts the curve.

Question 3 (Start a new page)

- (a) Simplify $\frac{1 + \tan^2 \theta}{1 + \cot^2 \theta}$ 2
- (b) ABCD is a parallelogram in which A is the point (4, 1). The equation of side BC is $3x - y + 5 = 0$ and the equation of side CD is $x + 2y + 2 = 0$. 5
- i) Find the equation of the side AD.
 - ii) Find the equation of the diagonal AC.
- (c) (Angles in this part may be left in decimal form, correct to 2 dec. places)
- i) Write $8\cos x + 6\sin x$ in the form $A\cos(x - \alpha)$ where α is acute. 2
 - ii) Hence solve $8\cos x + 6\sin x = 5$ for $0^\circ \leq x \leq 360^\circ$. 2

Question 4 (Start a new page)

(a) If $f(x) = \frac{1}{x}$

i) Write an expression for $f(a+h) - f(a)$ as a single fraction. 1

ii) Hence evaluate $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$. 2

iii) What does the answer to ii) represent? 1

iv) Describe briefly the graphical process which is represented 1
by the expression $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

(b) i) By expanding $\sin(x + 2x)$ and using the double angle formulas, 3
prove the identity $\sin 3x = 3 \sin x - 4 \sin^3 x$.

ii) Hence solve the equation $\sin 3x = 2 \sin x$ for $0^\circ \leq x \leq 360^\circ$. 3

Question 5 (Start a new page)

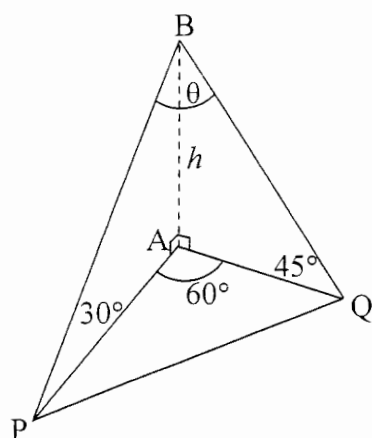
(a) i) Differentiate $y = \sqrt{x^2 + 1}$ 1

ii) Hence find the derivative of $y = x^2 \sqrt{x^2 + 1}$ as a fraction in simplest form. 2

(b) i) Express $\operatorname{cosec} \theta$ in terms of t , where $t = \tan \frac{\theta}{2}$. 1

ii) Hence show that $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$. 2

(c)



AB is a vertical tower of height h which stands on level ground. Angles of elevation of B from P and Q are 30° and 45° respectively. Angle PAQ is 60° and angle PBQ is θ .

i) Write PA and QA in terms of h 1

ii) By considering triangle PAQ show that $PQ^2 = (4 - \sqrt{3})h^2$ 1

iii) By finding a second expression for PQ^2 , show that 3

$$\cos \theta = \frac{2 + \sqrt{3}}{4\sqrt{2}}$$

QUESTION 1

a) i) $\lim_{x \rightarrow \infty} \frac{x^2 + 3x + 1}{2x^2 - 4}$

$= \lim_{x \rightarrow \infty} \frac{1 + \frac{3}{x} + \frac{1}{x^2}}{2 - \frac{4}{x^2}}$

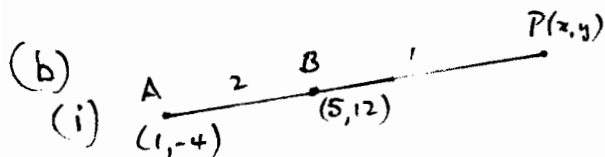
$= \frac{1}{2} \quad \checkmark$

ii) $\lim_{x \rightarrow 3} \frac{x^2 + 3x - 18}{x^2 - 5x + 6}$

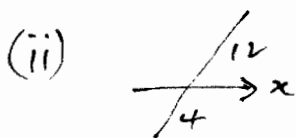
$= \lim_{x \rightarrow 3} \frac{(x+6)(x-3)}{(x-3)(x-2)}$

$= \lim_{x \rightarrow 3} \frac{(x+6)}{x-2}$

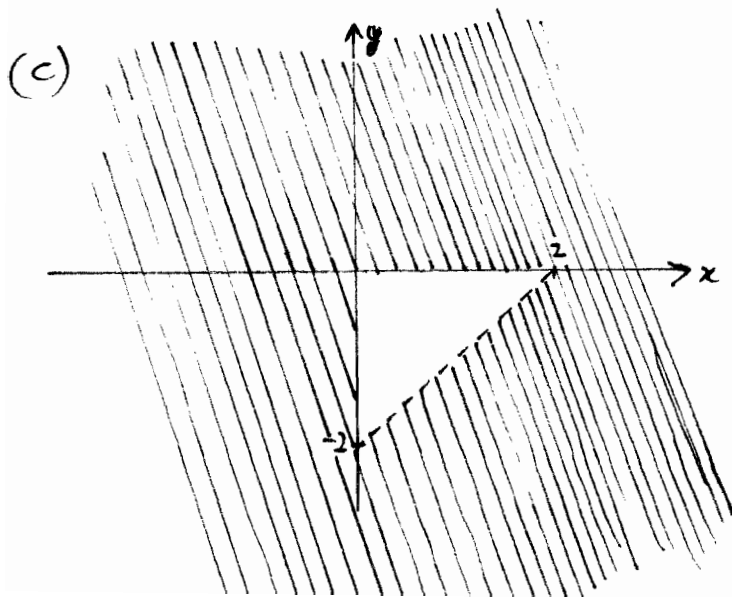
$= 9 \quad \checkmark$



P is $(7, 20)$



Ratio is 1:3.



a) i) Correct answer - 1 mark.

ii) Correct answer - 1 mark

b) (i) • Correct point - 2 marks
• x OR y value correct - 1 mark

• A correct substitution into the formula - 1 mark.

ii) Correct answer - 1 mark

(c)

Correct diagram - 3 marks.
(1 off each error)

$$(d) \quad y = 3x + 5, \quad y = mx + 4$$

$$m_1 = 3, \quad m_2 = m$$

$$\tan 45^\circ = \left| \frac{3 - m}{1 + 3m} \right| \quad \checkmark$$

$$\text{ie } \frac{3 - m}{1 + 3m} = 1 \quad \text{or} \quad \frac{3 - m}{1 + 3m} = -1$$

$$3 - m = 1 + 3m$$

$$\therefore m = 1/2 \quad \checkmark$$

$$3 - m = -1 - 3m$$

$$2m = -4$$

$$\therefore m = -2 \quad \checkmark$$

Correct values - 3 marks.

1 Correct value - 2 marks

Correct use of formula - 1 mark.

QUESTION 2

$$a) \quad i) \quad \frac{d}{dx} [f(x) + g(x)] = f'(x) + g'(x) \quad (1)$$

$$ii) \quad \frac{d}{dx} [f(x) \cdot g(x)] = f'(x)g(x) + f(x) \cdot g'(x) \quad (1)$$

$$iii) \quad \frac{d}{dx} \frac{1}{f(x)} = \frac{-f'(x)}{[f(x)]^2} \quad (1)$$

$$iv) \quad \frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x) \quad (1)$$

$$b) \quad y = \frac{x+1}{x-1}$$

$$i) \quad x = -1 \quad (1)$$

$$ii) \quad \frac{dy}{dx} = \frac{(x-1) - (x+1)}{(x-1)^2}$$

$$= \frac{-2}{(x-1)^2}$$

$$iii) \quad \text{At } P(x=-1), \quad m = -\frac{2}{4} = -\frac{1}{2} \quad (1)$$

Q2(cont)

(iv) Slope of Normal is 2 ✓
since $2 \times -\frac{1}{2} = -1$.

∴ at P(-1, 0)

$$y - 0 = 2(x + 1) \quad \checkmark$$

i.e. $y = 2x + 2$.

(v) When $2x + 2 = \frac{x+1}{x-1}$ ✓

$$2(x+1)(x-1) = x+1$$

$$\therefore x = -1, \frac{1}{2}$$

∴ Normal meets curve again when $x = \frac{1}{2}$. ✓

Correct working - 2 marks.

Use of point/gradient form must be shown.

Otherwise 1 mark for correctly stating gradient of normal.

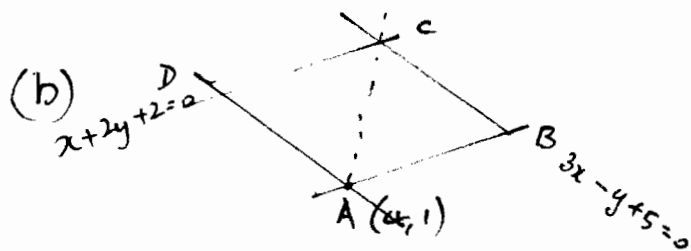
$x = \frac{1}{2}$ must explicitly be shown as the answer to this part. ("State clearly...")

① for initial equation

① for answer.

QUESTION 3

$$\begin{aligned} \text{a) } \frac{1 + \tan^2 \theta}{1 + \cot^2 \theta} &= \frac{\sec^2 \theta}{\operatorname{cosec}^2 \theta} \\ &= \tan^2 \theta \end{aligned}$$



i) Eqn AD:

$$3(x-4) - 1(y-1) = 0$$

$$\text{i.e. } 3x - y - 11 = 0$$

ii) Eqn AC is

$$x + 2y + 2 + k(3x - y + 5) = 0$$

and contains (4, 1)

$$\therefore 4 + 2 + 2 + k(12 - 1 + 5) = 0$$

$$8 + 11k = 0$$

Correct answer ② marks

Knowing 1 Pythagorean identity ① mark.

Correct answer ② marks.

Either correct gradient or ratio of coefficients ① mark.

✓ for set up

$\therefore$ Eqn AC is

$$2x + 4y + 4 - 3x + y - 5 = 0$$

$$\text{ie. } -x + 5y - 1 = 0$$

$$\text{or } x - 5y + 1 = 0$$

$$(c) i) 8 \cos x + 6 \sin x = A \cos(x - \alpha)$$

$$\therefore A \sin \alpha = 6$$

$$A \cos \alpha = 8$$

$$\therefore A = \sqrt{8^2 + 6^2} = 10$$

$$\alpha = \tan^{-1} \frac{3}{4} \div 36.87^\circ$$

$$\therefore 8 \cos x + 6 \sin x = 10 \cos(x - 36.87^\circ)$$

$$ii) 10 \cos(x - 36.87^\circ) = 5$$

$$\therefore \cos(x - 36.87^\circ) = \frac{1}{2}$$

$$\therefore x - 36.87^\circ = 60^\circ, 300^\circ$$

$$\therefore x \div 96.87^\circ, 336.87^\circ$$

Correct equation - (3) marks.

① for value of A

① for α

Correct answers (2) marks.

-1 for each error

QUESTION 4

$$(a) i) f(a+h) - f(a)$$

$$= \frac{1}{a+h} - \frac{1}{a}$$

$$= \frac{a - a - h}{a(a+h)}$$

$$= \frac{-h}{a(a+h)}$$

$$ii) \lim_{h \rightarrow 0} \frac{\frac{-h}{a(a+h)}}{h} = \lim_{h \rightarrow 0} \frac{-1}{a(a+h)}$$

$$= -1$$

① for correct answer.

① for dividing by h
or

(2) for correct answer

(iii) Gradient of the curve
at point where $x=a$.
OR Slope of (tangent to) the
curve at the point $(a, \frac{1}{a})$.

(iv) Gives the gradient of the
secant in its final position
as tangent to the curve.

ie Secant becomes tangent
as h becomes zero.

$$\begin{aligned} \text{b) i) } \sin(x+2x) &= \sin x \cos 2x + \cos x \sin 2x \\ &= \sin x (1 - 2\sin^2 x) + \cos x (2\cos x \sin x) \\ &= \sin x - 2\sin^3 x + 2\cos^2 x \sin x \\ &= \sin x - 2\sin^3 x + 2(1 - \sin^2 x) \sin x \\ &= \sin x - 2\sin^3 x + 2\sin x - 2\sin^3 x \\ &= 3\sin x - 4\sin^3 x \end{aligned}$$

$$\begin{aligned} \text{ii) } 3\sin x - 4\sin^3 x &= 2\sin x \\ \therefore 4\sin^3 x - \sin x &= 0 \\ \text{ie. } \sin x (4\sin^2 x - 1) &= 0 \\ \sin x (2\sin x - 1)(2\sin x + 1) &= 0 \\ \therefore \sin x &= 0, \frac{1}{2}, -\frac{1}{2} \\ \therefore x &= 0^\circ, 30^\circ, 150^\circ, 210^\circ, 330^\circ, \\ &\quad 180^\circ, 360^\circ \end{aligned}$$

Answer must refer to $x=a$
or the point where $x=a$. (1) mark.
Eg $(a, \frac{1}{a})$.

Simply stating "the derivative" or
"the slope" is in sufficient.

(1) mark - idea of tangent
as limiting position of
secant is required.

(b)

(1) for initial expansion.
(2) for showing another 2
correct substitutions - either
 $\sin 2x$ or $\cos 2x$ or
TOTAL (3) MARKS.

6 correct angles - (3) MARKS

IF $0^\circ, 360^\circ$ missing - (2) MARKS.

OR

Either finding $\sin x = \frac{1}{2}$ or
 $x = 30^\circ$ or (1) MARK.
equivalent

QUESTION 5

a) i) $y = (x^2 + 1)^{\frac{1}{2}}$
 $y' = \frac{1}{2}(x^2 + 1)^{-\frac{1}{2}} \times 2x$
 $= \frac{x}{\sqrt{x^2 + 1}}$ ✓

1 MARK.

ii) $y' = 2x\sqrt{x^2 + 1} + x^2 \cdot \frac{x}{\sqrt{x^2 + 1}}$
 $= \frac{2x(x^2 + 1) + x^3}{\sqrt{x^2 + 1}}$
 $= \frac{3x^3 + 2x}{\sqrt{x^2 + 1}}$

PRODUCT RULE CORRECT (1)

Correct answer (2) marks.

(b) i) $\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{1+t^2}{2t}$

(1) MARK.

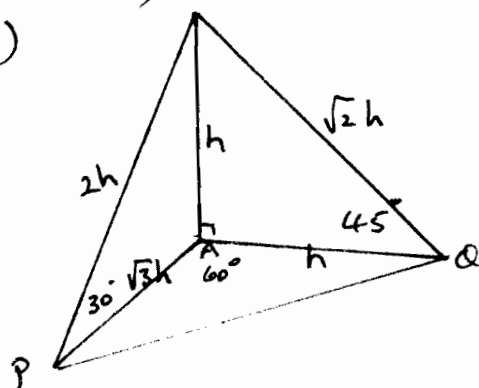
ii) LHS = $\operatorname{cosec} \theta + \cot \theta$
 $= \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$
 $= \frac{2}{2t}$
 $= \frac{1}{t}$
 $= \cot \frac{\theta}{2} = \text{RHS.}$

Correct substitution for $\cot \theta$ or equivalent (1) MARK.
 OR

Correct down to $\frac{1}{t}$ - (2) MARKS.

Q5(cont.) B

(c)



i) $PA = \sqrt{3}h$, $QA = h$

ii) Using cosine rule

$$PQ^2 = (\sqrt{3}h)^2 + h^2 - 2(\sqrt{3}h)(h) \cdot \cos 60^\circ$$

$$= 3h^2 + h^2 - \sqrt{3}h^2$$

$$= 4h^2 - \sqrt{3}h^2 \text{ as req'd.}$$

iii) In $\triangle PBQ$

$$PQ^2 = (2h)^2 + (\sqrt{2}h)^2 - 2 \cdot 2h \cdot \sqrt{2}h \cdot \cos \theta$$

$$= 4h^2 + 2h^2 - 4\sqrt{2}h^2 \cdot \cos \theta$$

$$= 6h^2 - 4\sqrt{2}h^2 \cdot \cos \theta$$

$$\therefore (4 - \sqrt{3})h^2 = 6h^2 - 4\sqrt{2}h^2 \cdot \cos \theta$$

$$\therefore \cos \theta = \frac{4 - \sqrt{3} - 6}{-4\sqrt{2}}$$

$$= \frac{2 + \sqrt{3}}{4\sqrt{2}} \text{ as req'd.}$$

① MARK - both correct

Cosine rule correct - ① mark.

PB and QB correct ① MARK.
OR

Correct expression for
 PQ^2 - ② MARKS
OR

Correct to here - ③ MARKS